C4.8 Write transactions

This section defines the state changes associated with the Write transaction group that are issued on the AW channel. The Write transactions are:

- *WriteNoSnoop*
- *WriteUnique* on page C4-203
- *WriteLineUnique* on page C4-203
- *WriteBack* on page C4-204
- *WriteClean* on page C4-204.

For details of the specific restrictions and the handling of overlapping writes see:

- *Restrictions on WriteUnique and WriteLineUnique Usage* on page C4-205
- *Handling overlapping write transactions* on page C4-207.

C4.8.1 WriteNoSnoop

A WriteNoSnoop transaction is used in a region of memory that is not shareable with other masters. A WriteNoSnoop transaction can result from:

- a program action, such as a store operation
- an update of main memory for a cache line that is in a non-shareable region of memory.

Table C4-20 shows the expected cache line state changes for the WriteNoSnoop transaction:

Table C4-20 Expected WriteNoSnoop cache line state changes

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteNoSnoop	I	I	I, UC, SC	I, UC, SC
	UC	UC	I, UC, SC	I, UC, SC
	UD	UC	I, UC, SC	I, UC, SC

Note

A cache line must only move from the Invalid state to a Valid state if a full cache line store has been performed.

Table C4-21 shows the other permitted cache line state changes for the WriteNoSnoop transaction:

Table C4-21 Other permitted WriteNoSnoop cache line state changes

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteNoSnoop	SC	UC	I, UC, SC	I, UC, SC
	SD	UC	I, UC, SC	I, UC, SC

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C4.8.2 WriteUnique

A WriteUnique transaction is used in a region of memory that is shareable with other masters. A single write occurs that is required to propagate to main memory.

There are restrictions on the use of WriteUnique transactions by cached masters that can hold dirty cache lines. See *Restrictions on WriteUnique and WriteLineUnique Usage* on page C4-205.

Table C4-22 shows the expected cache line state changes for the WriteUnique transaction:

Table C4-22 Expected WriteUnique cache line state changes

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteUnique	I	I	I	I
	UC	SC	SC	I, SC
	SC	SC	SC	I, SC

In the case of master holding a line in a Clean state while performing a WriteUnique transaction, the cache line must be updated to the new value when the WriteUnique transaction response is received.

C4.8.3 WriteLineUnique

A WriteLineUnique transaction is used in a region of memory that is shareable with other masters. A single write occurs, that is required to propagate to main memory.

Note

A WriteLineUnique transaction must be a full cache line store and all bytes within the cache line must be updated.

There are restrictions on the use of WriteLineUnique transactions by cached masters that can hold dirty cache lines. See *Restrictions on WriteUnique and WriteLineUnique Usage* on page C4-205.

Table C4-23 shows the expected cache line state changes for the WriteLineUnique transaction.

Table C4-23 Expected WriteLineUnique cache line state changes

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteLineUnique	I	I	I	I
	UC	SC	SC	I, SC
	SC	SC	SC	I, SC

In the case of master holding a line in a Clean state while performing a WriteLineUnique transaction, the cache line must be updated to the new value when the WriteLineUnique transaction response is received.

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C4.8.4 WriteBack

A WriteBack transaction is a write that can be used in shareable and non-shareable regions of memory. A WriteBack transaction is a write of a dirty cache line to update main memory.

Note

The difference between a WriteBack and a WriteClean transaction is whether or not the cache line remains allocated in the cache for a shareable region of memory. After a WriteBack transaction the cache line is no longer allocated. After a WriteClean transaction the cache line remains allocated.

The permitted state changes that Table C4-24 and Table C4-25 show, do not take into account a preceding store operation that makes a cache line dirty. If a store operation and WriteBack transaction occur as an atomic process, then the legal cache line state changes can be determined by combining the legal state changes for a store operation, see *State changes associated with a coherent store* on page C4-188, followed by the legal state changes for a WriteBack transaction.

Table C4-24 shows the expected cache line state changes for the WriteBack transaction in a shareable memory region.

Table C4-24 Expected WriteBack cache line state changes in a shareable memory region

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteBack	UD	I	I	I, UC, SC
	SD	I	I	I, SC

Table C4-25 shows the expected cache line state changes for the WriteBack transaction in a non-shareable memory region.

Table C4-25 Expected WriteBack cache line state changes in a non-shareable memory region

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteBack	UD	I	I, UC, SC	I, UC, SC
	SD	I	I, UC, SC	I, UC, SC

C4.8.5 WriteClean

A WriteClean transaction is a write operation that can be used in shareable and non-shareable regions of memory. A WriteClean transaction is a write of a dirty cache line to update main memory.

Note

The difference between a WriteClean and a WriteBack transaction is whether or not the cache line remains allocated in the cache for a shareable region of memory. After a WriteClean transaction the cache line remains allocated. After a WriteBack transaction the cache line is no longer allocated.

The permitted state changes that Table C4-26 on page C4-205 and Table C4-27 on page C4-205 show, do not take into account any preceding store operation that makes a cache line dirty. If a store operation and WriteBack transaction occur as an atomic process, then the legal cache line state changes can be determined by combining the legal state changes for a store operation, see *State changes associated with a coherent store* on page C4-188, followed by the legal state changes for a WriteClean transaction.

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Table C4-26 shows the expected cache line state changes for the WriteClean transaction in a shareable memory region.

Table C4-26 Expected WriteClean cache line state changes in a shareable memory region

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteClean	UD	UC	UC, SC	I, UC, SC
	SD	SC	SC	I, SC

Table C4-27 shows the expected cache line state changes for the WriteClean transaction in a non-shareable memory region.

Table C4-27 Expected WriteClean cache line state changes in a non-shareable memory region

Transaction	Start state	Expected end state	Legal end state	
			With Snoop Filter	No Snoop Filter
WriteClean	UD	UC	I, UC, SC	I, UC, SC
	SD	UC	I, UC, SC	I, UC, SC

C4.8.6 Restrictions on WriteUnique and WriteLineUnique Usage

Typically, WriteUnique and WriteLineUnique transactions are used by a non-cached component that is writing to a shareable region of memory. However, WriteUnique and WriteLineUnique transactions can be used by a cached component that meets the necessary requirements.

A cached component must be able to complete any incoming snoop transaction while a WriteUnique or WriteLineUnique transaction is in progress. This means that a cached component must:

- Complete any outstanding WriteBack or WriteClean transactions before issuing a WriteUnique or WriteLineUnique transaction.

  Note

  No additional WriteBack or WriteClean transactions can be issued until all outstanding WriteUnique or WriteLineUnique transactions are completed.

- Complete any incoming snoop transactions without the use of WriteBack or WriteClean transactions while a WriteUnique or WriteLineUnique transaction is in progress.

This is necessary, because earlier transactions, that also might require earlier snoop transactions to complete, can prevent WriteUnique and WriteLineUnique transactions from progressing.

These requirements restrict the use of WriteUnique and WriteLineUnique transactions to components that can either:

- complete all snoop transactions without requiring any data to be supplied, for example write-through caches that do not keep dirty cache lines for shareable data
- complete snoop transactions through the use of the snoop data channel, **CDDATA**.

These mechanisms only need to be used when a WriteUnique or WriteLineUnique transaction is in progress.

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